

PHYSICS

SECTION-1

1) A lamp radiates power P_0 uniformly in all directions, the amplitude of electric field strength E_0 at a distance r from it is-

(1) $E_0 = \frac{P_0}{2\pi\epsilon_0 cr^2}$

(2) $E_0 = \sqrt{\left\{ \frac{P_0}{2\pi\epsilon_0 cr^2} \right\}}$

(3) $E_0 = \sqrt{\left\{ \frac{P_0}{4\pi\epsilon_0 cr^2} \right\}}$

(4) $E_0 = \sqrt{\left\{ \frac{P_0}{8\pi\epsilon_0 cr} \right\}}$

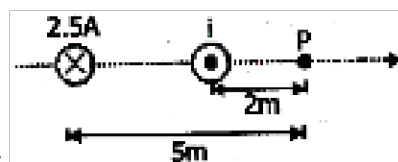
2) A current of 10 amp is flowing in a wire of length 1.5 m. A force of 15 newton acts on it when it is placed in a uniform magnetic field of 2 tesla. The angle between the magnetic field and the direction of current is -

(1) 30°

(2) 45°

(3) 60°

(4) 90°



3) The value of i so that magnetic field at P is zero-

(1) 1A

(2) 2A

(3) 3A

(4) 4A

4) A charged particle (q, m) is projected into a transverse uniform magnetic field B , with speed v . Regarding motion of the charged particle, quantities are listed in column-I and expressions are listed in column-II. Match the columns

	Column-I (Quantities)		Column-I (Expressions)
(A)	Radius of circular path (r)	(P)	$\frac{qB}{m}$
(B)	Angular velocity (ω) of	(Q)	$\frac{2\pi m}{qB}$
(C)	Time period (T) of circular motion	(R)	$\frac{mv}{qB}$

Select the correct option-

- (1) (A) - R, (B) - Q, (C) - P
- (2) (A) - P, (B) - R, (C) - Q
- (3) (A) - R, (B) - P, (C) - Q
- (4) (A) - Q, (B) - P, (C) - R

5) A long straight wire of length 2 m and mass 250 g is suspended horizontally in a uniform horizontal magnetic field of 0.7 T. The amount of current flowing through the wire will be ($g = 9.8 \text{ ms}^{-2}$)

- (1) 2.45 A
- (2) 2.25 A
- (3) 2.75 A
- (4) 1.75 A

6) An iron rod of susceptibility 599 is subjected to a magnetising field of 1200 Am^{-1} . The permeability of the material of the rod is ($\mu_0 = 4\pi \times 10^{-7} \text{ TmA}^{-1}$)

- (1) $2.4\pi \times 10^{-4} \text{ TmA}^{-1}$
- (2) $8.0\pi \times 10^{-5} \text{ TmA}^{-1}$
- (3) $2.4\pi \times 10^{-5} \text{ TmA}^{-1}$
- (4) $2.4\pi \times 10^{-7} \text{ TmA}^{-1}$

7) A point P lies on the axis of a flat coil carrying a current. The magnetic moment of the coil is μ . What will be the magnetic field at point P? It is given that the distance of P from the center of coil is d, which is large compared to the radius of the coil.

- (1) $\frac{\mu_0}{2\pi} \left(\frac{\mu}{d^3} \right)$

- (2) $\frac{\mu_0}{4\pi} \left(\frac{\mu}{d^3} \right)$

- (3) $\frac{\mu_0}{6\pi} \left(\frac{\mu}{d^2} \right)$

(4) $\frac{\mu_0}{8\pi} \left(\frac{\mu}{d^2} \right)$

8) Current in a circuit falls from 5.0 A to 0 A in 0.1 s. If an average emf of 200 V is induced, estimate the self-inductance of the circuit :-

- (1) 3 H
- (2) 2 H
- (3) 4 H
- (4) 5 H

9) In a step-up transformer the turn ratio is 1 : 10. A resistance of 200 ohm connected across the secondary is drawing a current of 0.5 A. What is the primary voltage and current ?

- (1) 20 V, 2 A
- (2) 25 V, 4 A
- (3) 10 V, 5 A
- (4) 50 V, 1 A

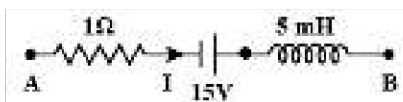
10) A coil of N turns and mean cross-sectional area A is rotating with uniform angular velocity ω about an axis at right angle to uniform magnetic field B. How much e.m.f. E will be induced in the coil?

- (1) $NBA\omega \sin \omega t$
- (2) $\frac{NB}{A} \sin \omega t$
- (3) $NB\omega \sin \omega t$
- (4) $NBA \sin \omega t$

11) A rod of length ℓ rotates with a uniform angular velocity ω about an axis passing through its middle point but normal to its length in a uniform magnetic field of induction B with its direction parallel to the axis of rotation. What is the induced emf between the two ends of the rod?

- (1) $\frac{B\ell^2\omega}{2}$
- (2) Zero
- (3) $2B\ell^2\omega$
- (4) $\left(\frac{B\ell^2\omega}{8} \right)$

12) The network shown in the following figure is a part of a complete circuit. If at a certain instant, the current I is 5 A and is decreasing at the rate of 10^3 A/s, then $V_B - V_A$ is



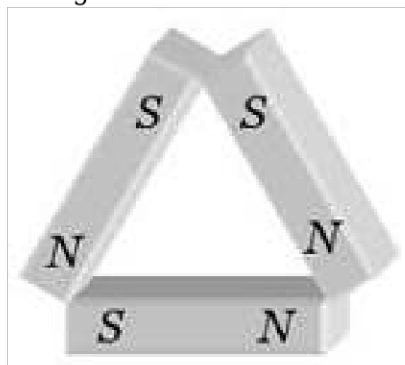
- (1) 20 V

- (2) 15 V
 (3) 10 V
 (4) 5 V

13) A copper disc of radius 0.1 m is rotated about its center with 10 revolutions per second in a uniform magnetic field of 0.1 tesla with its plane perpendicular to the field. How much e.m.f. is induced across the radius of disc ?

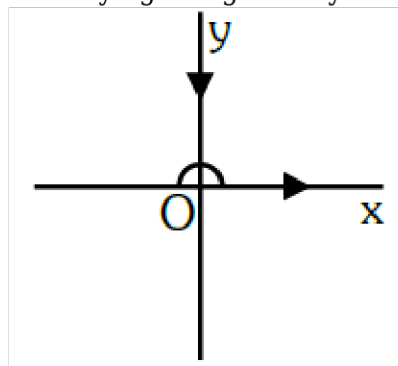
- (1) $2\pi \times 10^{-2} \text{V}$
 (2) $\pi \times 10^{-2} \text{V}$
 (3) $\frac{2\pi}{10} \text{V}$
 (4) $\frac{\pi}{10} \text{V}$

14) 3 identical bar magnets each of magnetic moment M are kept in the form of an equilateral triangle as shown below. What is the net magnetic moment of the system?



- (1) $\frac{3M}{2}$
 (2) $M\sqrt{3}$
 (3) $2M$
 (4) Zero

15) Equal currents are flowing in two infinitely long separate wires lying along x and y-axes in the



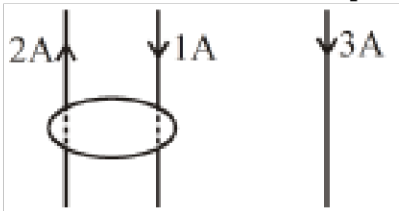
directions shown in figure. Match the following two columns.

Column-I	Column-II
(a) Magnetic field at (a, a)	(p) along positive y-axis

(b) Magnetic field at $(-a, -a)$	(q) along positive z-axis
(c) Magnetic field at $(a, -a)$	(r) along negative z-axis
(d) Magnetic field at $(-a, a)$	(s) zero

- (1) a-p, b-q, c-s, d-r
 (2) a-q, b-r, c-s, d-s
 (3) a-r, b-s, c-p, d-r
 (4) a-s, b-p, c-q, d-s

16) Two wires with currents 2 A and 1A are enclosed in circular loop. Another wire with current 3 A is situated outside the loop as shown. The $\oint \vec{B} \cdot d\vec{\ell}$ around the loop is :-



- (1) μ_0
 (2) $3\mu_0$
 (3) $6\mu_0$
 (4) $2\mu_0$

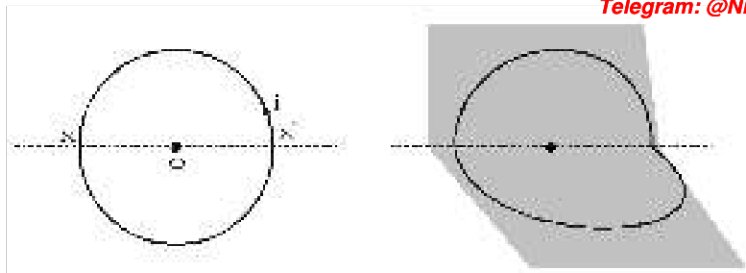
17) A cell is connected between the points A and C of a circular conductor ABCD of centre 'O'. $\angle AOC = 60^\circ$. If B_1 and B_2 are the magnitudes of the magnetic fields at O due to the currents in ABC



and ADC respectively, the ratio $\frac{B_1}{B_2}$ is

- (1) 6
 (2) 1
 (3) 5
 (4) $\frac{1}{5}$

18) In the given figure magnetic field at the center of ring (O) is $8\sqrt{2}T$. Now it is turned through a 90° about axis, so that two semicircular parts are mutually perpendicular. Then find the value of



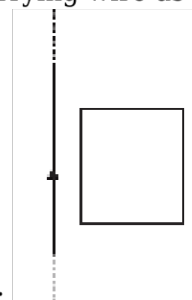
magnetic field (in Tesla) at centre.

- (1) 8 T
- (2) 16 T
- (3) 4 T
- (4) 12 T

19) The magnetic moment produced in a substance of 1 gm is 6×10^{-7} ampere - metre². If its density is 5 gm/cm^3 , then the intensity of magnetisation in A/m will be

- (1) 8.3×10^6
- (2) 3.0
- (3) 1.2×10^{-7}
- (4) 3×10^{-6}

20) A square loop of conducting wire is placed near a long straight current carrying wire as shown.



Match the statements in column-I with the corresponding results in column-II:

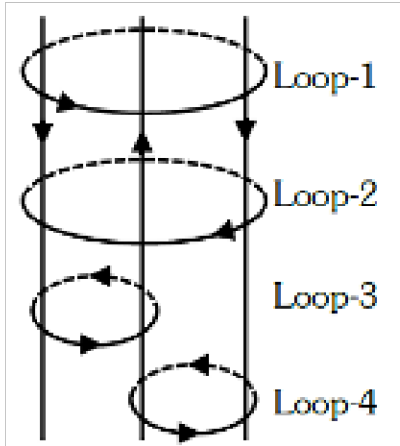
Column-I	Column-II
(A) If the magnitude of current I is increased	(p) Induced current in the loop will be clockwise.
(B) If the magnitude of current I is decreased	(q) Induced current in the loop will be anticlockwise.
(C) If the loop is moved away from the wire	(r) Wire will attract the loop
(D) If the loop is moved towards the wire	(s) Wire will repel the loop

- (1) A(q, s), B(p, r), C(p, r), D(q, s)
- (2) A(r, q), B(r, p), C(q, p), D(r, p)
- (3) A(p, q), B(r, p), C(q, s), D(r, p)
- (4) None of these

21) The magnitude flux through each turn of a 100 turn coil is $(t^3 - 2t) \times 10^{-3} \text{ Wb}$, where t is in second. The induced emf at $t = 2 \text{ s}$ is

- (1) -4 V
- (2) -1 V
- (3) +1 V
- (4) +4 V

22) Three wires are carrying same constant current i in different directions. Four loops enclosing the wires in different manners are shown in figure. The direction of $d\vec{\ell}$ is shown in the figure.



Column-I	Column-II
(a) Along close Loop-1	(p) $\oint \vec{B} \cdot d\vec{\ell} = \mu_0 i$
(b) Along close Loop-2	(q) $\oint \vec{B} \cdot d\vec{\ell} = -\mu_0 i$
(c) Along close Loop-3	(r) $\oint \vec{B} \cdot d\vec{\ell} = 0$
(d) Along close Loop-4	(s) $\oint \vec{B} \cdot d\vec{\ell} = 2\mu_0 i$

- (1) a - q, b - p, c - r, d - r
- (2) a - p, b - r, c - s, d - s
- (3) a - r, b - p, c - s, d - p
- (4) a - r, b - q, c - q, d - q

23) A charged particle revolves in circular path in uniform transverse magnetic field after being accelerated by a potential difference of V volt. If V is doubled, then choose the correct options.

- (a) Kinetic energy of particle will become two times.
- (b) Radius of circular path will become two times.
- (c) Radius of circular path will become $\sqrt{2}$ times.
- (d) angular velocity will remain unchanged.

- (1) (a) and (b) are correct
- (2) (a) and (c) are correct

- (3) (a), (c), (d) are correct
 (4) None

24) In an L-R circuit, the value of L is $(0.4/\pi)$ henry and the value of R is 30 ohm. If in the circuit, an alternating emf of 200 volt at 50 cycles per second is connected, the impedance of the circuit and effective current will be:

- (1) 11.4 ohm, 17.5 ampere
 (2) 30.7 ohm, 6.5 ampere
 (3) 40.4 ohm, 5 ampere
 (4) 50 ohm, 4 ampere.

25) The potential difference across an instrument in an a.c. circuit of frequency f is V and the current flowing through it is I such that $V = 5 \cos(2\pi ft)$ volt and $I = 2 \sin(2\pi ft)$ amp. The average power dissipated in the instrument is:

- (1) Zero
 (2) 10 watt
 (3) 5 watt
 (4) 2.5 watt

26) A coil of resistance 200 ohms and self inductance 1.0 henry has been connected to an a.c. source of frequency $\frac{200}{\pi}$ Hz. The phase difference between voltage and current is:

- (1) 30°
 (2) 63°
 (3) 45°
 (4) 90°

27) The value of L, C and R in an LCR series circuit are 4 mH, 40 pF and 100Ω respectively. The quality factor of the circuit is

- (1) 10,000
 (2) 100
 (3) 1000
 (4) 10

28) The ratio of magnetic inductions at the centre of a circular coil of radius a and on its axis at a distance equal to its radius, will be -

- (1) $\frac{1}{\sqrt{2}}$
 (2) $\frac{\sqrt{2}}{1}$
 (3) $\frac{1}{2\sqrt{2}}$

(4) $\frac{2\sqrt{2}}{1}$

29) Match the column quantities in column-I with their units in column-II.

Column-I		Column-II	
(i)	Magnetic field	(a)	Henry/meter
(ii)	Magnetic permeability	(b)	A/m
(iii)	Magnetising intensity	(c)	Henry
(iv)	Self inductance	(d)	Weber/m ²

- (1) (i)-d, (ii)-a, (iii)-b, (iv)-c
 (2) (i)-b, (ii)-c, (iii)-b, (iv)-a
 (3) (i)-b,d, (ii)-a, (iii)-a, (iv)-a
 (4) (i)-b, (ii)-c, (iii)-a, (iv)-d

30) A rectangular coil of length 0.12 m and width 0.1 m having 50 turns of wire is suspended vertically in a uniform magnetic field of strength 0.2 Weber/m². The coil carries a current of 2 A. If the plane of the coil is inclined at an angle of 30° with the direction of the field, the torque required to keep the coil in stable equilibrium will be :

- (1) 0.12 Nm
 (2) 0.15 Nm
 (3) 0.20 Nm
 (4) 0.24 Nm

31) The magnetic susceptibility of a paramagnetic material at -73°C is 0.0075 then its value at -173°C will be :-

- (1) 0.0045
 (2) 0.0030
 (3) 0.015
 (4) 0.0075

32) A proton, deuteron and an α -particle are accelerated by same potential, enter in uniform magnetic field perpendicularly. Ratio of radii of circular path respectively :-

- (1) $1 : \sqrt{2} : \sqrt{2}$
 (2) 2 : 2 : 1
 (3) 1 : 2 : 1
 (4) 1 : 1 : 1

33) A radiation of energy E falls normally on a perfectly reflecting surface. The momentum transferred to the surface is:

- (1) E/c

- (2) $2E/c$
 (3) Ec
 (4) E/c^2

34) Match List-I with List-II

	List-I (Electromagnetic waves)		List-II (Wavelength)
(a)	AM radio wave	(i)	10^{-10} m
(b)	Microwaves	(ii)	10^2 m
(c)	Infrared radiation	(iii)	10^{-2} m
(d)	X-rays	(iv)	10^{-4} m

Choose the correct answer from the options given below:-

- (1) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)
 (2) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)
 (3) (a)-(iii), (b)-(ii), (c)-(i), (d)-(iv)
 (4) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)

35) Electromagnetic radiation propagating in free space has electrical average energy density (U_e) and magnetic average energy density (U_m)

- (1) $U_e = U_m$
 (2) $U_e > U_m$
 (3) $U_e < U_m$
 (4) $U_e \neq U_m$

SECTION-2

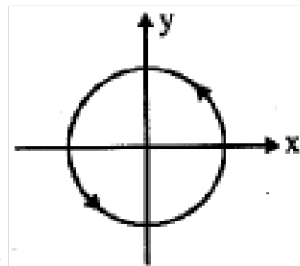
1) Given below are two statements:

Statement I: Electromagnetic waves carry energy as they travel through space and this energy is equally shared by the electric and magnetic fields.

Statement II: When electromagnetic waves strike a surface, a pressure is exerted on the surface. In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is incorrect but Statement II is correct
 (2) Both Statement I and Statement II are correct.
 (3) Both Statement I and Statement II are incorrect.
 (4) Statement I is correct but Statement II is incorrect.

2) A circular current carrying loop is placed in x-y plane as shown in figure. A uniform magnetic field

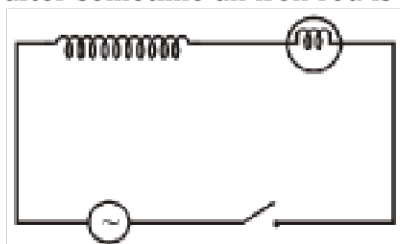


$\vec{B} = B_0 \hat{k}$ is present in the region. Match the following-

	Table-I		Table-II
(A)	Magnetic moment of the loop	(p)	Zero
(B)	Torque on the loop	(q)	Maximum
(C)	Potential energy of the loop	(r)	Along positive z-axis
(D)	Equilibrium of the loop	(s)	Stable
		(t)	Minimum

- (1) A - (r), B - (p), C - (t), D - (s)
 (2) A - (r), B - (p), C - (q), D - (s)
 (3) A - (r), B - (p), C - (s), D - (q)
 (4) A - (q), B - (r), C - (p), D - (s)

3) A light bulb and an inductor coil are connected to an ac source through a key as shown in the figure below. The key is closed and after sometime an iron rod is inserted into the interior of the



inductor. The glow of the light bulb

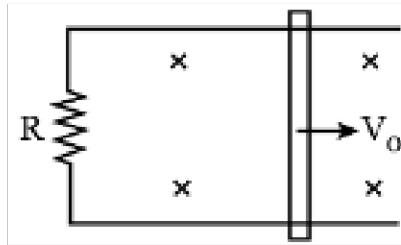
- (1) decreases
 (2) remains unchanged
 (3) will fluctuate
 (4) increases

4) Voltage and current in an AC circuit are given by and

$$V = 5 \sin \left(100\pi t - \frac{\pi}{6} \right) \quad I = 4 \sin \left(100\pi t + \frac{\pi}{6} \right)$$

- (1) Voltage leads the current by 30°
 (2) Current leads the voltage by 30°
 (3) Current leads the voltage by 60°
 (4) Voltage leads the current by 60°

5) In the given figure conductor AB is given initial horizontal velocity v_0 then correct statement for



its subsequent motion.

- (1) Its velocity will remain constant
- (2) Its velocity increases
- (3) Its velocity decreases

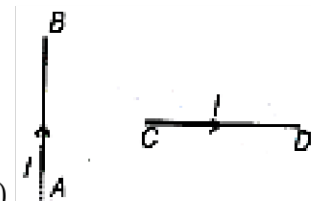
- (4) It executes SHM with the frequency $\omega = \frac{B\ell}{\sqrt{ML}}$

6) A constant current I maintained in a solenoid. Which of the following quantities will increase if an iron rod is inserted in the solenoid along its axis:-

- (a) Magnetic field at the centre
- (b) Magnetic flux linked with the solenoid
- (c) Self inductance of the solenoid
- (d) Rate of Joule heating

- (1) a, b, c
- (2) c, d
- (3) a, b
- (4) Only b

7) Two current carrying conductors are placed on a smooth surface as shown in figure, then for conductor CD.

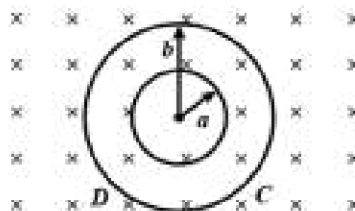


(where a and α are acceleration and angular acceleration respectively)

- (1) $a = 0, \alpha = 0$
- (2) $a \neq 0, \alpha \neq 0$
- (3) $a = 0, \alpha \neq 0$
- (4) $a \neq 0, \alpha = 0$

8) Plane figures made of thin wires of resistance $R = 50$ milli ohm/meter are located in a uniform magnetic field perpendicular into the plane of the figures and which decrease at the rate

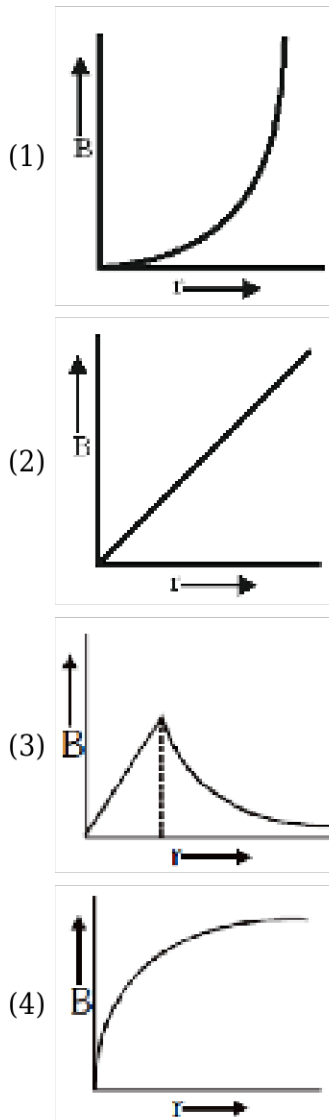
$\frac{dB}{dt} = 0.1 \text{ m T/s}$. Then what are the respective currents in the inner and outer boundary? (The inner



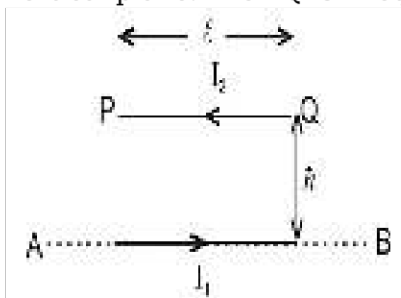
radius $a = 10 \text{ cm}$ and outer radius $b = 20 \text{ cm}$)

- (1) 2×10^{-4} A (anticlockwise), 10^{-4} A (anticlockwise)
 (2) 2×10^{-4} A (clockwise), 10^{-4} A (anticlockwise)
 (3) 10^{-4} A (anticlockwise), 2×10^{-4} A (clockwise)
 (4) 10^{-4} A (clockwise), 2×10^{-4} A (clockwise)

9) A thick current carrying cable of radius 'R' carries current 'I' uniformly distributed across its cross-section. The variation of magnetic field $B(r)$ due to the cable with the distance 'r' from the axis of the cable is represented by :



10) In the figure shown the wires AB and PQ carry constant currents I_1 and I_2 respectively. PQ is of uniformly distributed mass 'm' and length ' ℓ '. AB and PQ are both horizontal and kept in the same vertical plane. The PQ is in equilibrium at height 'h' where h is equal to



- (1) $\frac{\mu_0 I_1 I_2 \ell}{\pi m g}$
 (2) $\frac{\mu_0 I_1 I_2}{2 \pi m g}$
 (3) $\frac{\mu_0 I_1 I_2 \ell}{2 \pi m g}$
 (4) $\frac{\mu_0 I_1 I_2 \ell}{2 \pi}$

11) Magnetic flux in a circular coil of resistance 10Ω changes with time as shown in figure. Symbol $\otimes$ indicates a direction perpendicular to paper inwards. Match the following :

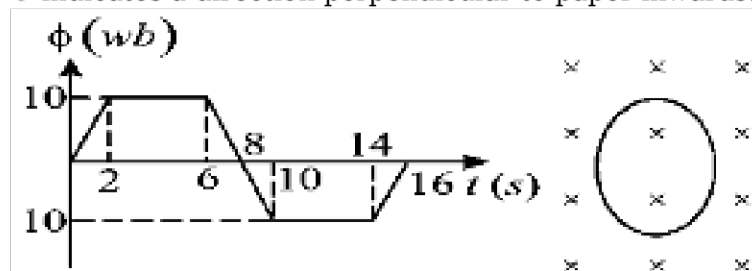


Table-1	Table-2
(A) At 1 s induced current is	(p) Clockwise
(B) At 5 s induced current is	(q) Anticlockwise
(C) At 9 s induced current is	(r) Zero
(D) At 15 s induced current is	(s) 2A
	(t) None

- (1) A-q; B-r; C-p; D-q
 (2) A-p; B-r; C-q; D-p
 (3) A-r; B-p; C-q; D-q
 (4) A-p; B-r; C-s; D-q

12) A bar magnet of moment M is cut into two identical pieces along the length. One piece is bent in the form of a semi circle. If two pieces are perpendicular to each other, then resultant magnetic moment is

- (1) $\left(\frac{M}{\pi}\right)^2 + \left(\frac{M}{2}\right)^2$
 (2) $\sqrt{\left(\frac{M}{\pi}\right)^2 + \left(\frac{M}{2}\right)^2}$
 (3) $\sqrt{\left(\frac{M}{\pi}\right)^2 - \left(\frac{M}{2}\right)^2}$

(4) $\frac{M}{\pi} + \frac{M}{2}$

13) Match the column I with column II -

	Column-I(Law)		Column-II(Mathematical formula)
(A)	Faraday law	(P)	$\oint \vec{E} \cdot d\vec{A} = \frac{q}{\epsilon_0}$
(B)	Ampere Maxwell law	(Q)	$\oint \vec{B} \cdot d\vec{A} = 0$
(C)	Gauss law for magnetism	(R)	$\oint \vec{E} \cdot d\vec{\ell} = -\frac{d\phi_B}{dt}$
(D)	Gauss law for electrostatics	(S)	$\oint \vec{B} \cdot d\vec{\ell} = \mu_0 i + \epsilon_0 \mu_0 \frac{d\phi_E}{dt}$

- (1) A → R, B → Q, C → S, D → P
 (2) A → P, B → Q, C → S, D → R
 (3) A → P, B → S, C → Q, D → R
 (4) A → R, B → S, C → Q, D → P

14) The electric field associated with an electromagnetic wave in vacuum is given by

$\vec{E} = 40 \cos(kz - 6 \times 10^8 t) \hat{i}$, where E, z and t are in volt per meter, meter and second respectively.

The value of wave vector k is :-

- (1) 2 m^{-1}
 (2) 0.5 m^{-1}
 (3) 6 m^{-1}
 (4) 3 m^{-1}

15) Match the following.

	Column-I		Column-II
(a)	Resonant frequency	(i)	$V.I. \cos\theta$
(b)	Average power	(ii)	$\sqrt{R^2 + (X_L - X_C)^2}$
(c)	impedence	(iii)	$\frac{1}{R} \sqrt{\frac{L}{C}}$
(d)	Quality factor	(iv)	$\frac{1}{2\pi\sqrt{LC}}$

- (1) a-iv, b-iii, c-ii, d-i
 (2) a-iv, b-i, c-ii, d-iii
 (3) a-i, b-ii, c-iii, d-iv
 (4) a-i, b-iii, c-ii, d-iv

1)

The rate constant and rate of reaction are same for :

- (1) First order
- (2) Zero order
- (3) Second order
- (4) n^{th} order

2)

In the reaction, $2A + B \rightarrow \text{Product}$, when concentration of B alone was doubled, the half life did not change. When the concentration of A alone was doubled, the rate increases by two times. The unit of rate constant of the reaction is

- (1) s^{-1}
- (2) $\text{L mol}^{-1} \text{s}^{-1}$
- (3) Unitless
- (4) $\text{mol L}^{-1} \text{s}^{-1}$

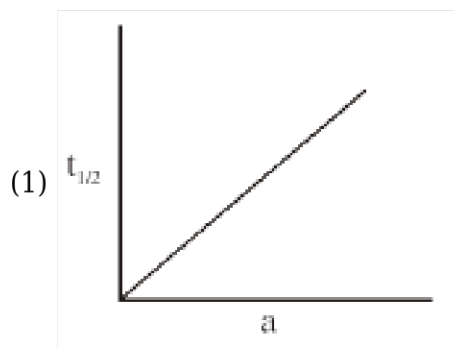
3)

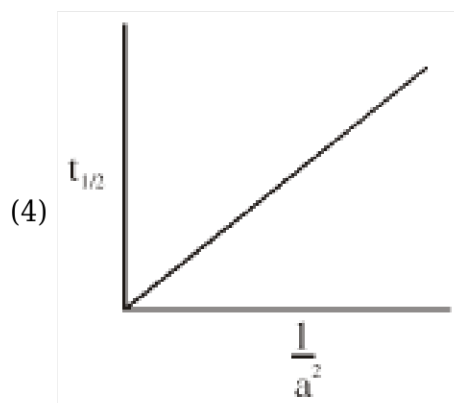
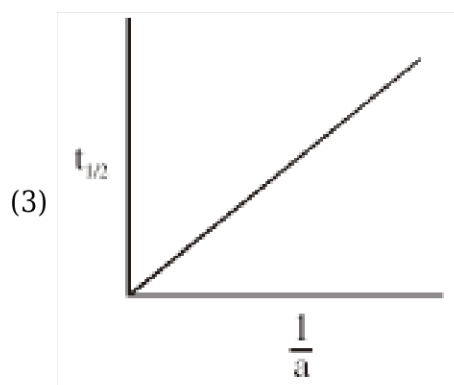
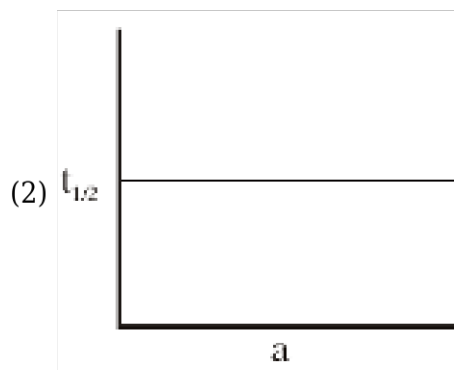
The rate of the elementary reaction, $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$, when the volume of the reaction vessel is doubled :

- (1) Eight times of its initial rate
- (2) One-eighth of its initial rate
- (3) Four times of its initial rate
- (4) One-fourth of its initial rate

4)

Which of the following curve represents a first order reaction?





5) A first order reaction is 50% complete in 20 min, the time taken for 99.9% completion is :-

- (1) 30 min
- (2) 200 min
- (3) 50 min
- (4) 38 min

6) The rate constant, the activation energy and the Arrhenius parameter of a chemical reaction at 25°C are $3 \times 10^{-4} \text{ s}^{-1}$, $104.4 \text{ kJ mol}^{-1}$ and $6 \times 10^{14} \text{ s}^{-1}$ respectively. The value of the rate constant as $T \rightarrow \infty$ is

- (1) $2 \times 10^{18} \text{ s}^{-1}$
- (2) $6 \times 10^{14} \text{ s}^{-1}$
- (3) ∞
- (4) $3.6 \times 10^{30} \text{ s}^{-1}$

7) The activation energy of exothermic reaction, $A \rightarrow B$ is 80 kJ mol^{-1} . The heat of reaction is 200 kJ mol^{-1} . The activation energy of the reaction $B \rightarrow A$

(in kJ/mol) will be

- (1) 120
- (2) 140
- (3) 280
- (4) 200

8) What is the rate equation for a reaction,
 $A + B \rightarrow \text{products}$, based on the rate data ?

$[A]_0, \text{mol L}^{-1}$	$[B]_0, \text{mol L}^{-1}$	Rate
0.15	0.10	x
0.30	0.20	4x
0.30	0.40	16x

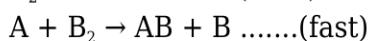
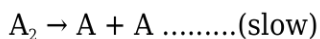
- (1) $\text{Rate} = k[A]^2$
- (2) $\text{Rate} = k[B]^2$
- (3) $\text{Rate} = k[A][B]$
- (4) $\text{Rate} = k[A][B]^2$

9) The temperature coefficient of a reaction is:-

- (1) The rate constant
- (2) The rate constant at a fixed temperature
- (3) The ratio of rate constant at two temperature
- (4) The ratio of rate constant at two temperature differing by 10°C preferably k_{308}/k_{298}

10)

$A_2 + B_2 \rightarrow 2AB$ occurs by the following mechanism -



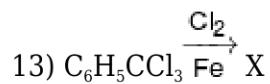
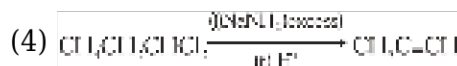
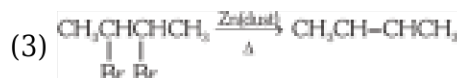
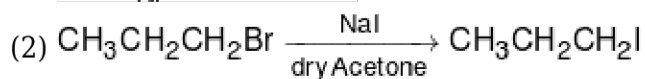
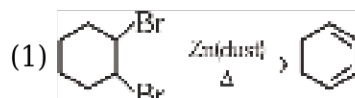
its order would be -

- (1) $3/2$
- (2) 1
- (3) 0
- (4) 2

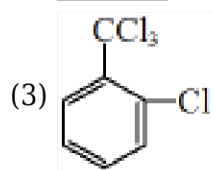
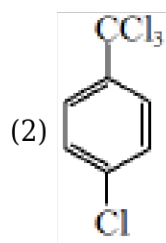
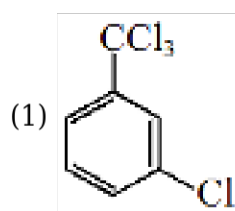
11) Which of the following has the highest boiling point

- (1) $\text{CH}_3\text{CH}_2\text{I}$
- (2) CH_3Cl
- (3) CH_3I
- (4) CH_3Br

12) Which reaction product is wrong (major) product

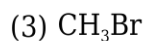


In the above reaction X is

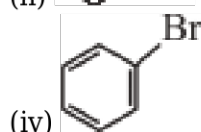
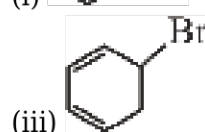
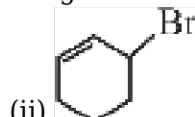
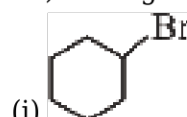


(4) All in equal amount

14) Which of the following has highest dipole moment :



15) Arrange the following in order of ease of dehydrohalogenation:



(1) (iii) > (iv) > (ii) > (i)

(2) (iii) > (ii) > (i) > (iv)

(3) (ii) > (iii) > (i) > (iv)

(4) (i) > (ii) > (iii) > (iv)

16)

Which one is most reactive towards S_N1 reaction?

(1) $C_6H_5CH_2Br$

(2) $C_6H_5CH(C_6H_5)Br$

(3) $C_6H_5CH(CH_3)Br$

(4) $C_6H_5C(CH_3)(C_6H_5)Br$

17)

$MeCH_2C\equiv CH \xrightarrow{NH_3/NaNH_2} A \xrightarrow{EtBr} B$,
A and B are

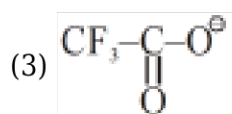
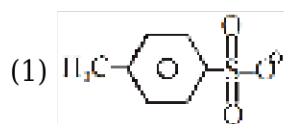
(1) $MeCH_2C\equiv CNa$, $MeCH_2C\equiv C-Et$

(2) $MeCH_2CH=CH_2$, $MeCH_2-CH(Et)-CH_3$

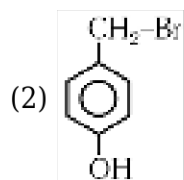
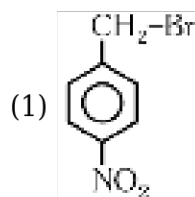
(3) $MeCH_2CH=CHNH_2$,
 $MeCH_2CH=CH-NHBr$

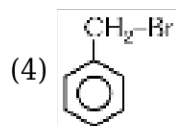
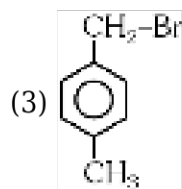
(4) $MeCH_2C\equiv C-NH_2$, $MeC\equiv C-NH-Br$

18) Which of the following has maximum nucleophilicity :-



19) Which is most reactive for S_N1 ?



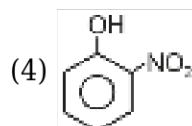
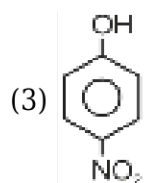


20) Alkyl fluorides are synthesized by

- (1) Finkelstein reaction
- (2) Swarts reaction
- (3) Kolbe reaction
- (4) Wurtz reaction

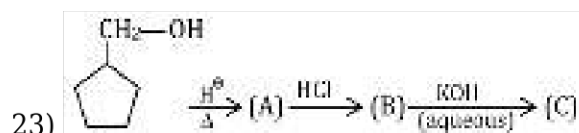
21) Which among the following compounds do not react with sodium metal?

- (1) CH_3OH
- (2) CH_3OCH_3

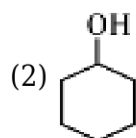
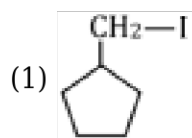


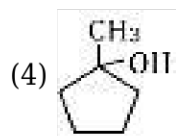
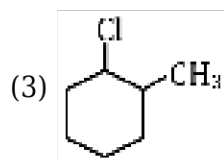
22) Phenyl magnesium bromide reacts with methanol to give

- (1) A mixture of anisole and $\text{Mg}(\text{OH})\text{Br}$
- (2) A mixture of benzene and $\text{Mg}(\text{OMe})\text{Br}$
- (3) A mixture of toluene and $\text{Mg}(\text{OH})\text{Br}$
- (4) A mixture of phenol and $\text{Mg}(\text{Me})\text{Br}$

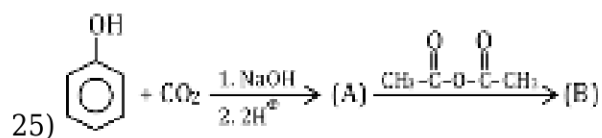
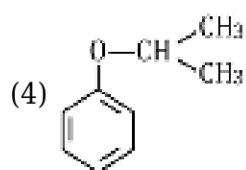
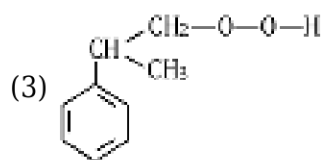
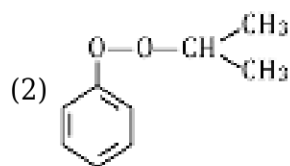
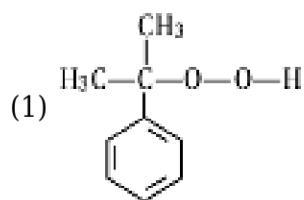
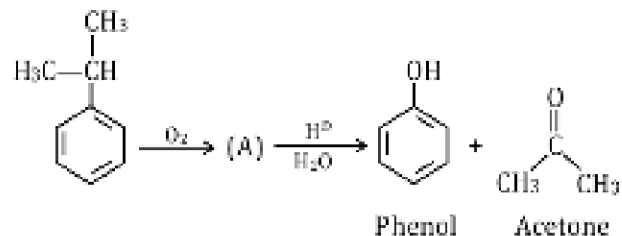


What will be "C" ?

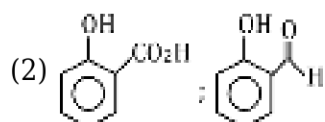
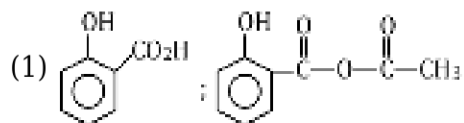


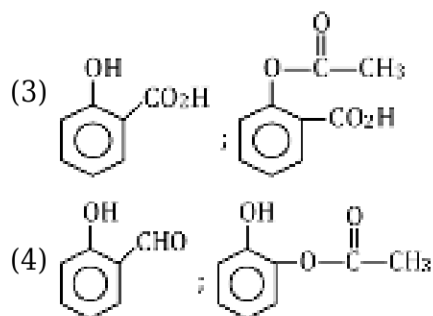


24) The structure of intermediate A in the following reaction is :-



Product (A) and (B) respectively are :-



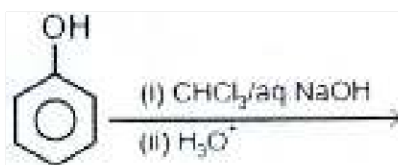


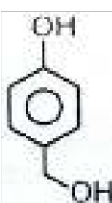
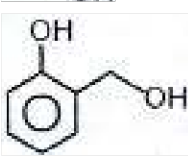
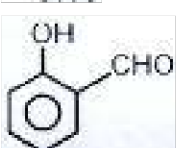
26) **Assertion:** p-nitrophenol has high pK_a in comparison to o-nitrophenol.

Reason: In o-nitrophenol, intermolecular H-bonding is present.

- (1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
- (2) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion
- (3) Assertion is false but Reason is true.
- (4) Assertion is true but Reason is false.

27) Major product of the given reaction is :

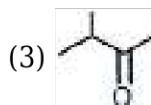


- (1) 
- (2) 
- (3) 
- (4) 

28) The major product of the given reaction is :



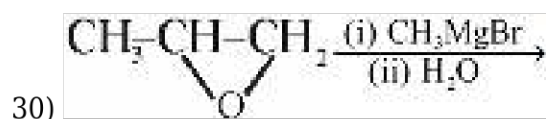
- (1) 
- (2) 



29) **Assertion :** Phenol is more reactive than benzene towards electrophilic substitution reaction

Reason : In the case of phenol, the intermediate carbocation is not resonance stabilized.

- (1) Assertion is true, Reason is true and Reason is correct explanation for Assertion
- (2) Assertion is true, Reason is true and Reason is not correct explanation for Assertion
- (3) Assertion is true, Reason is false
- (4) Assertion is false, Reason is true (or) Both Assertion & Reason are false



- (1) $\text{CH}_3\text{-CH}(\text{CH}_3)\text{-CH}_2\text{-OH}$
- (2) $\text{CH}_3\text{-CH}(\text{OH})\text{-CH}_2\text{-CH}_3$
- (3) $\text{CH}_3\text{-CH}(\text{CH}_3)\text{-CH}_3$
- (4) $\text{HO-CH}_2\text{-CH}_2\text{-CH}_2\text{-CH}_2\text{-OH}$

31) Trisilylamine $(\text{SiH}_3)_3\text{N}$ is :

- (1) Trigonal pyramidal and strong acidic
- (2) Trigonal pyramidal and weak acidic
- (3) Trigonal planar and strong basic
- (4) Trigonal planar and weak basic

32) Which of the following species show minimum melting and boiling point :

- (1) s-block elements
- (2) d-block elements
- (3) Inert gas
- (4) All p-block elements

33) Which of the following species act as strong reducing agent :

- (1) NH_3
- (2) PH_3
- (3) BiH_3
- (4) AsH_3

34) Which of the following species is liquid due to hydrogen bonding:

- (1) HCl

- (2) HBr
- (3) HF
- (4) HI

35) CO_2 gas & SO_2 gas are distinguished by using

- (1) NaOH
- (2) NH_3
- (3) $\text{K}_2\text{Cr}_2\text{O}_7$
- (4) HCl

SECTION-2

1) Which of the following species give pure N_2 gas on Heating ?

- (1) NH_4NO_2
- (2) $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$
- (3) NaN_3
- (4) All

2) **Statement-1** : Acid strength increases in the order given as $\text{HF} < \text{HCl} < \text{HBr} < \text{HI}$.

Statement-2 : As the size of the elements F, Cl, Br, I increases down the group, the bond strength of HF, HCl HBr and HI decreases and so the acid strength increases.

- (1) If both statements are true and the statement-2 is correct explanation of statement-1.
- (2) If both statements are true but statement-2 is not correct explanation of statement-1.
- (3) If statement-1 is true but the statement-2 is false.
- (4) If statement-1 is false but the statement-2 is true

3) The molecule shows more vapour pressure at room temperature

- (1) H_2O
- (2) H_2S
- (3) H_2Te
- (4) H_2Se

4) Which statement is/are true for SN^2 reaction :- I. The rate of reaction is independent of the concentration of the nucleophile.

II. The nucleophile attacks the carbon atom on the side of the molecule, opposite to the group being displaced.

III. The reaction process with simultaneous bond formation and bond rupture.

- (1) I, II
- (2) I, III
- (3) I, II, III

(4) II, III

5) Least reactive towards N.S.R (Nucleophilic Substitution Reaction) :-

(1) $\text{CH}_2=\text{CH}-\text{CH}_2-\text{Cl}$

(2) CH_3-Cl



(1) 1

(2) 2

(3) 3

(4) 4

6) Pick up the correct statement about alkyl halides:

(1) They show H-bonding

(2) They are soluble in water

(3) They are soluble in organic solvent

(4) They do not contain any polar bond

7) $\text{S}_{\text{N}}2$ reaction involve the formation of

(1) Carbocation

(2) Carbanion

(3) Free radical

(4) Transition state

8) Alcoholic solution of KOH is used for

(1) Dehydration

(2) Dehydrogenation

(3) Dehydrohalogenation

(4) Dehalogenation

9) Which of the following have smallest carbon-halogen bond length

(1) $\text{C}-\text{F}$

(2) $\text{C}-\text{Cl}$

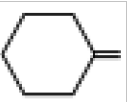
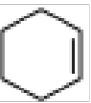
(3) $\text{C}-\text{Br}$

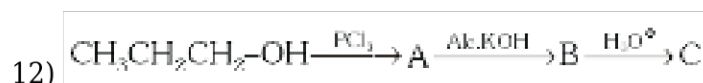
(4) $\text{C}-\text{I}$

10) The reaction of sodium ethoxide with ethyl iodide to form diethyl ether is termed as

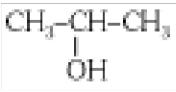
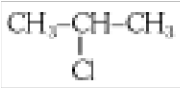
- (1) Electrophilic Substitution
- (2) Nucleophilic Substitution
- (3) Electrophilic Addition
- (4) Radical Substitution

11) Hydroboration-oxidation, oxymercuration-demercuration and acid catalysed hydration will give same product in :-

- (1) 
- (2) 
- (3) 
- (4) 



Find product 'C' is

- (1) $\text{CH}_3\text{CH=CH}_2$
- (2) 
- (3) 
- (4) $\text{CH}_3\text{CH}_2\text{CH}_2\text{-Cl}$

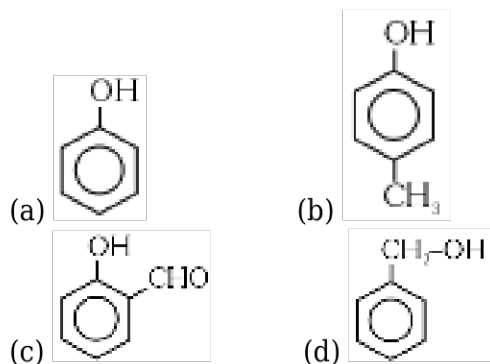
13) Which of the following compounds will give butanone on oxidation with alkaline KMnO_4 solution ?

- (1) Butan-1-ol
- (2) Butan-2-ol
- (3) Both of these
- (4) None of these

14) Which of the following compounds will react with sodium hydroxide solution in water ?

- (1) $\text{C}_6\text{H}_5\text{OH}$
- (2) $\text{C}_6\text{H}_5\text{CH}_2\text{OH}$
- (3) $(\text{CH}_3)_3\text{COH}$
- (4) $\text{C}_2\text{H}_5\text{OH}$

15) Which of the following will gives characteristic colour with FeCl_3 ?

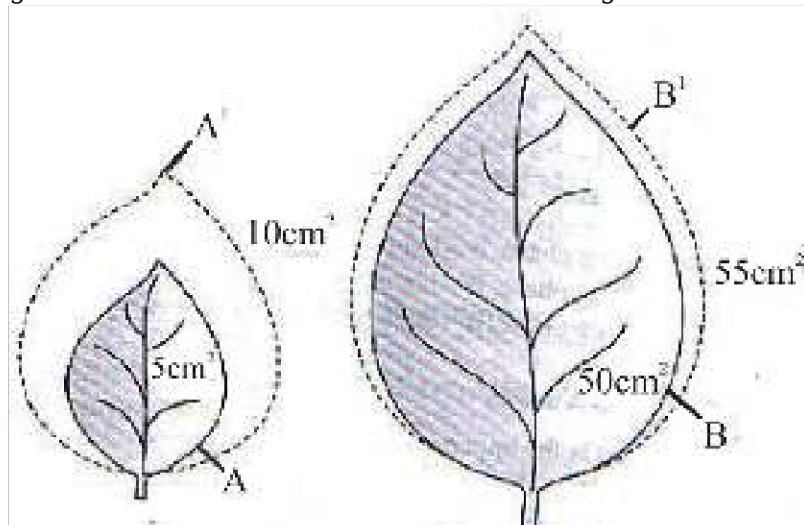


- (1) a, b, c, d
 (2) a, b, c
 (3) only a
 (4) a & b

BOTANY

SECTION-1

1) Two leaves, A and B are of different sizes but shows absolute increase in area in the unit time to give leaves A^1 and B^1 . Find out the absolute growth and relative growth rate of both the leaves :



	Absolute growth of A	Absolute growth of B	Relative growth rate of A	Relative growth rate of B
(1)	5 cm ²	10 cm ²	10%	50%
(2)	10 cm ²	5 cm ²	5%	10%
(3)	5 cm ²	5 cm ²	10%	100%
(4)	5 cm ²	5 cm ²	100%	10%

- (1) 1
 (2) 2

(3) 3

(4) 4

2) The effect of apical dominance can be overcome by which of the following hormone

(1) IAA

(2) Ethylene

(3) Gibberellin

(4) Cytokinin

3) Enhances the respiration rate during ripening of the fruits. This rise in the rate of respiration is called:

(1) Respiratory climax

(2) Respiratory rise

(3) Respiratory climactic

(4) None of these

4) Auxins are isolated by-

(1) Charles Darwin

(2) Francis Darwin

(3) F W. Went

(4) Both (1) and (2)

5) Match the following for PGR.

Column I	Column II
A. IAA	— + i. Carotenoids
B. Kinetin	— + ii. Terpenes
C. ABA	— + iii. Indole compounds
D. GA ₃	— + iv. Adenine derivatives

(1) A-iii, B-iv, C-i, D-ii

(2) A-iv, B-ii, C-iii, D-i

(3) A-i, B-iii, C-ii, D-iv

(4) A-ii, B-i, C-iv, D-iii

6) The ability of plants to follow different pathways in response to environment or phases of life to form different kind of structures is called?

(1) Adaptation

(2) Differentiation

(3) Maturation

(4) Plasticity

7) All those changes which an organism goes through during its life cycle from germination of the

seed to senescence is termed as-

- (1) Growth
- (2) Differentiation
- (3) Development
- (4) None of these

8) Geometric growth graphically shows

- (1) Linear curve
- (2) Sigmoid curve
- (3) Both one by one
- (4) Straight line

9) What is re-differentiation?

- (1) Regaining the capacity to divide
- (2) Loosing the capacity to divide
- (3) Loosing the capacity to divide after regaining
- (4) All of these

10) By how many times, the cells of watermelon increase in size?

- (1) 315000
- (2) 350000
- (3) 325000
- (4) 317000

11) Increased vacuolation, cell enlargement and new cell wall deposition are the characteristics of cell in :-

- (1) Divisional phase
- (2) Enlongation phase
- (3) Maturation phase
- (4) Differentiation phase

12) To promote the growth of lateral branches of a plant:

- (1) Axillary buds are removed
- (2) Auxin is applied to apical bud
- (3) Auxin is applied to the decapitated shoot tip
- (4) Apical bud is removed

13) In which of these plants, heterophyllous development occurs due to the environment?

- (1) Coriander

- (2) Cotton
- (3) Larkspur
- (4) Buttercup

14) Which one of the following generally acts as an antagonist to gibberellins?

- (1) Zeatin
- (2) Ethylene
- (3) ABA
- (4) IAA

15) "Foolish seedling" disease of rice led to the discovery of

- (1) ABA
- (2) 2, 4-D
- (3) IAA
- (4) GA

16) Aquatic plant which is pollinated by insects or wind

- (1) *Zostera*
- (2) Water hyacinth
- (3) *Vallisneria*
- (4) *Amorphophallus*

17) Diameter of pollen grains are generally about:

- (1) 10-25 micrometer
- (2) 25-50 micrometer
- (3) 50-150 micrometer
- (4) 150-300 micrometer

18) Egg apparatus consists of

- (1) One egg cell + Antipodal + Synergids
- (2) One egg cell + Two Synergids
- (3) Antipodals + Central cell
- (4) One egg cell + Central cell

19) How many megaspore mother cells are required to produce 100 eggs during meiosis:

- (1) 1
- (2) 100
- (3) 50
- (4) 25

20) How many male gametes can be produced by 4 pollen mother cells?

- (1) 8
- (2) 32
- (3) 64
- (4) 16

21) Intine of pollen grains is composed of:

- (1) Pectocellulose
- (2) Lignocellulose
- (3) Sporopollenin
- (4) Pollen kitt

22) In a dry and dormant seed the moisture content is mostly around

- (1) 0%
- (2) 0-5%
- (3) 5-10%
- (4) 10-15%

23) Stalk of the ovule is called

- (1) Hilum
- (2) Funicle
- (3) Integument
- (4) Chalaza

24) More than one nucleus is found in which wall layer of anther:

- (1) Epidermis
- (2) Tapetum
- (3) Middle Layer
- (4) Endothecium

25) The pollination which brings genetically different type of pollen grains to the stigma is.....

- (1) Autogamy
- (2) Geitonogamy
- (3) Xenogeny
- (4) Cleistogamy

26) In which of the following plant releases pollen grains on the surface of water during pollination?

- (1) *Vallisnaria*
- (2) *Zostera*

- (3) *Hydrilla*
 (4) Water hyacinth

27) Match **Correctly**.

	Column I		Column II
(a)	Central Cell	(i)	Megasporangia
(b)	Ovule	(ii)	Pollen Allergy
(c)	Carrot grass	(iii)	Remnants of nucellus
(d)	Perisperm	(iv)	Polar nuclei

- (1) a-ii, b-iv, c-i, d-iii
 (2) a-iii, b-i, c-ii, d-iv
 (3) a-iv, b-i, c-ii, d-iii
 (4) a-iv, b-ii, c-i, d-iii

28) Albuminous seeds are found in

- (1) Wheat and maize
 (2) Castor and barley
 (3) Pea and groundnut
 (4) Both 1 and 2

29) False fruits are found in

- (1) Bananas and grapes
 (2) Tomato and guava
 (3) Strawberry and cashew
 (4) Papaya and orange

30) Outbreeding devices are used to prevent -

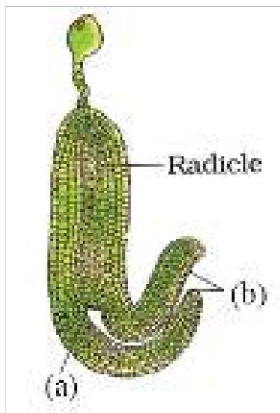
- (1) Self pollination
 (2) Cross-pollination
 (3) Both self and cross pollination
 (4) Xenogamy

31) In grass family, the scutellum is -

- (1) Cotyledon
 (2) Root tip
 (3) Epiblast
 (4) Shot tip

32) Triple fusion is -

- (1) Fusion of third male gamete with polar nuclei
- (2) Fusion of three haploid cells
- (3) Fusion of second male gamete with egg cell
- (4) Fusion of one male gamete and two polar nuclei



33)

Identify (a) and (b) in given diagram.

- (1) (a) Hypocotyl (b) root cap.
- (2) (a) Plumule (b) Endosperm
- (3) (a) Coleoptile (b) Cotyledons
- (4) (a) Plumule (b) Cotyledons

34) The coconut water from tender coconut represents:-

- (1) Endocarp
- (2) Fleshy mesocarp
- (3) Free nuclear proembryo
- (4) Free nuclear endosperm

35) Coconut milk contains

- (1) ABA
- (2) Auxin
- (3) Zeatin
- (4) Gibberellin

SECTION-2

1) Growth can be measured in various ways. Which of these can be used as parameters to measure growth?

- (1) Increase in cell number
- (2) Increase in cell size
- (3) Increase in length and weight
- (4) All the above

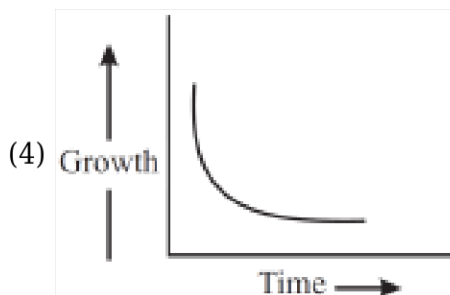
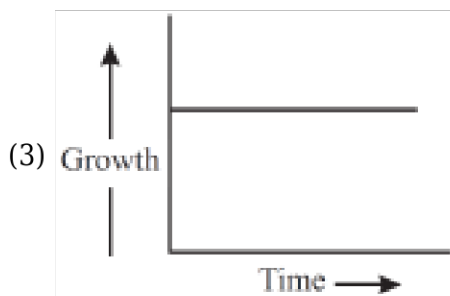
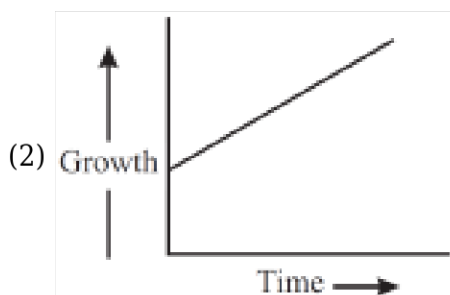
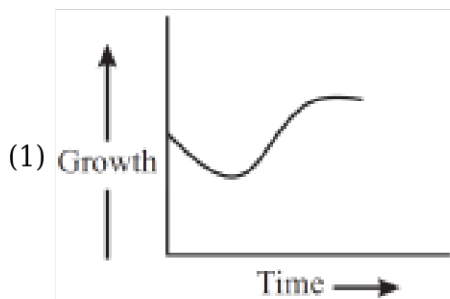
2) The factors affecting growth can be:

- (1) Water, light, gravity but not temperature
- (2) Water, temperature, light but not gravity
- (3) Water, temperature, light and gravity
- (4) Water, temperature, gravity but not light

3) Maximal size in terms of wall thickening and protoplasmic modification are achieved by :-

- (1) Cells of divisional phase
- (2) Cells of maturation phase
- (3) Cells of elongation phase
- (4) Cells of meristematic tissue

4) Which of the following shows the **correct** graph of arithmetic growth?



5) Study the following statements.

I	"X" hormone promotes root growth and root hair formation thus helping the plants to increase their absorption surface.
II	"Y" hormone induces flowering in mango and also promotes rapid internode/petiole elongation in deep plants and hence helping leaves or upper parts of shoot to remain above water.
III	"Z" hormone inhibits the seed germination, increase the tolerance of plant to various stresses, play import in seed development, maturation and dormancy.

Identify the **correct** names of hormones marked as 'X', 'Y' & 'Z'.

- (1) Y = ABA; X = Auxin; Z = GA
- (2) Z = GA; X = Auxin; Y = Ethylene
- (3) Y = Auxin; X = Ethylene; Z = GA
- (4) Y = Ethylene; X = Ethylene; Z = ABA

6) The living differentiated cells, that by now have lost the capacity to divide can regain the capacity of division under certain conditions is called

- (1) Undifferentiation
- (2) Redifferentiation
- (3) Dedifferentiation
- (4) Differentiation

7) Match the **column A** and **Column B**

Column A		Column B	
(a)	Vegetative cell	(i)	Small
(b)	Generative cell	(ii)	Abundant reserve food
		(iii)	Spindle shaped
		(iv)	Bigger
		(v)	Irregular shape of nucleus

- (1) a-ii, iv, v; b-i, iii
- (2) a-i, ii,v; b-iii, iv
- (3) a-i, iii, b-ii, iv, v
- (4) a-i, iv, v b-ii, iii

8) What will be the ploidy of endosperm and zygote respectively. If the cross is made between 6n female and 4n male plant

- (1) 8n, 8n
- (2) 8n, 5n
- (3) 10n, 7n
- (4) 6n, 4n

9) Which of the following statements are **correct**?

- (i) Some species of *Asteraceae* and grasses produce seeds without fertilization.
- (ii) In apomixis, diploid egg may develop in to embryo without fertilization.
- (iii) In mango, nucellar cells form embryo directly
- (iv) In Polyembryony, multiple embryos are not formed in seeds.

- (1) ii, iii, iv
- (2) i, ii, iii
- (3) i , ii , iv
- (4) i, iii, iv

10) Match **Correctly**

	Column I		Column II
(a)	<i>Yucca</i> flower	(i)	Wind Pollination
(b)	<i>Zostera</i>	(ii)	Safe place to lay eggs
(c)	Corn Tassel	(iii)	Assured seed set
(d)	<i>Oxalis</i>	(iv)	Marine sea grass

- (1) a-ii , b-iv, c-i, d-iii
- (2) a-ii, b-ii, c-iv, d-ii
- (3) a-iv, b-ii, c-i, d-iii
- (4) a-ii, b-iv, c-iii, d-i

11) **Statement I** : Hybrid varieties of several of our food and vegetable crops are being extensively cultivated.

Statement II : Cultivation of hybrids has tremendously increased productivity.

- (1) Only statement I is correct
- (2) Only statement II is correct
- (3) Both statements I and II are correct
- (4) Both statements I and II are incorrect

12) **Assertion:** The pistil has the ability to recognize the pollen, whether it is of the right type or of the wrong type.

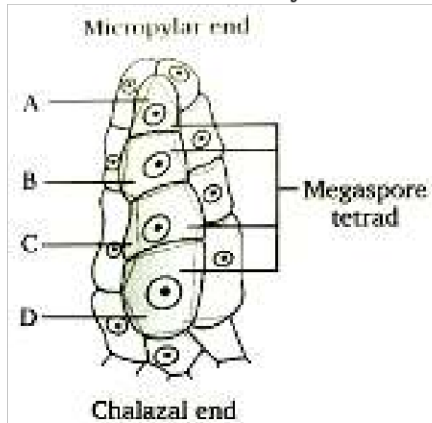
Reason: Pollination does guarantee the transfer of the right type of pollen.

- (1) Both Assertion (A) and Reason (R) are the true and Reason (R) is a correct explanation of Assertion (A).
- (2) Both Assertion (A) and Reason (R) are the true but Reason (R) is not a correct explanation of Assertion (A).
- (3) Assertion (A) is true and Reason (R) is false.
- (4) Both Assertion (A) and Reason (R) are false.

13) Choose the **correct** match regarding the development of flower into fruit.

- (1) Wall of ovule - pericarp
- (2) Nucellus - periderm
- (3) Ovary - seed
- (4) Ovary wall - fruit wall

14) Observe the four megaspores of an ovule depicted below and find out which of these (A, B, C, D) survive to form embryo sac and which of these (A, B, C, D) degenerates?



	Survives	Degenerates
1	B, C, D	A
2	D	A, B, C
3	A, B, C	D
4	A	B, C, D

- (1) 1
- (2) 2
- (3) 3
- (4) 4

15) Match the following:

(A) 2, 4-D	(i) Herring sperm DNA
(B) ABA	(ii) Bolting
(C) Ethylene	(iii) Stomatal closure
(D) GA	(iv) Weed-free lawns
(E) Cytokinins	(v) Ripening of fruits

- (1) (A) - (iv), (B) - (iii), (C) - (v), (D) - (ii), (E) - (i)
- (2) (A) - (v), (B) - (iii), (C) - (iv), (D) - (ii), (E) - (i)
- (3) (A) - (iv), (B) - (i), (C) - (v), (D) - (iii), (E) - (ii)
- (4) (A) - (v), (B) - (iii), (C) - (ii), (D) - (i), (E) - (iv)

ZOOLOGY

SECTION-1

1) Which is not a natural method of contraception ?

- (1) Coitus interruptus
- (2) Periodic abstinence
- (3) Vasectomy
- (4) Lactational amenorrhea

2) Which one of the following is not the function of placenta ?

- (1) Secretes estrogen
- (2) Facilitates removal of carbon dioxide and waste material from embryo
- (3) Secretes oxytocin during parturition
- (4) Facilitates supply of oxygen and nutrients to embryo

3) Name the ART method, where embryos with more than 8 blastomeres are transferred into the uterus?

- (1) ICSI
- (2) IUT
- (3) IUI
- (4) AI

4) are also essential in certain cases where continuation of pregnancy could be harmful or even fatal either to the mother or to the foetus or both:-

- (1) MTPs
- (2) IUDs
- (3) Contraceptive pill
- (4) Female condom

5) Name the contraceptive drug prepared by central drug research institute in Lucknow?

- (1) Progestasert
- (2) Saheli
- (3) Mala D
- (4) Mala L

6) Which one of the following does indicate Improved Reproductive Health of Society?

- (a) Better Detection and Cure of STDs.
- (b) Increase number of medically assisted deliveries.
- (c) Increases number of people with small families.
- (d) Better awareness about sex related matters

- (1) (a) and (b)
- (2) (b), (c) and (d)
- (3) (a), (c) and (d)

(4) (a), (b), (c) and (d)

7) Progestasert is a

- (1) Oral contraceptive
- (2) Natural contraceptive
- (3) Hormonal IUD
- (4) Implant contraceptive

8) **Lippes loop** is a

- (1) Structure associated with hormone releasing IUDs
- (2) Structure associated with male reproductive system
- (3) Structure associated with ligamentous tissue
- (4) A non-medicated IUDs

9) Complications of STDs lead to further complications like

- (1) Abortion
- (2) Still birth
- (3) Ectopic pregnancy
- (4) All of the above

10) The age group of _____ years is quite vulnerable to STDs.

- (1) 10-19
- (2) 15-22
- (3) 17-27
- (4) 15-24

11) Specialised procedure to form embryo in lab in which sperm is directly injected into the ovum is ?

- (1) AI
- (2) IUT
- (3) IUI
- (4) ICSI

12) **Statement-I** : In vasectomy, small part of vas deferens is removed or tied through a small incision on the scrotum.

Statement-II : Surgical methods are highly effective but their reversibility is very poor.

- (1) Statement I and Statement II are correct.
- (2) Statement I and Statement II are incorrect.
- (3) Statement I is correct but Statement II is incorrect.
- (4) Statement I is incorrect but Statement II is correct.

13) Which one is true for the oral contraceptive "SAHELI" ?

- (1) It is once a week pill
- (2) It has multiple side effects
- (3) It is non-steroidal with high contraceptive value
- (4) Both (1) & (3)

14) The aim of **artificial Insemination (AI)** technique is used in which of the following conditions-

- (1) The female who is unable to produce fertile egg
- (2) The female whose endometrial lining not synchronized with ovulation
- (3) The female where the husband is either unable to inseminate or has very low sperm count
- (4) The female who has undergoes tubal ligation

15) Which of the following method used for determining condition of foetus for genetic counselling -

- (1) MTP
- (2) Amniocentesis
- (3) Southern blotting
- (4) Zoo blotting

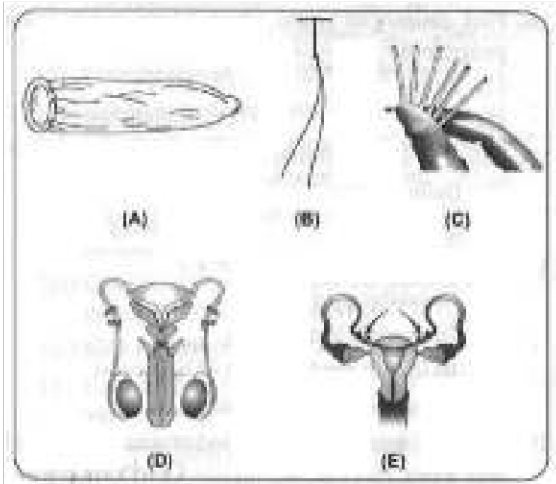
16) Which of the following sexually transmitted diseases is not completely curable even at early stage?

- (1) Gonorrhoea
- (2) Genital warts
- (3) Genital herpes
- (4) Chlamydiasis

17) Days for periodic abstinence are :

- (1) 10th - 11th day of menstrual cycle
- (2) 10th - 17th day of menstrual cycle
- (3) 17th - 20th day of menstrual cycle
- (4) 20th - 28th day of menstrual cycle

18) Match the following figures to the options given below and choose the correct answer.



- (I) Tubectomy
(II) Vasectomy
(III) Implants
(IV) Condom
(V) IUD

- (1) A - IV, B - V, C - III, D - II, E - I
(2) A - IV, B - II, C - II, D - I, E - V
(3) A - I, B - II, C - III, D - IV, E - V
(4) A - III, B - IV, C - V, D - II, E - I

19) Emergency contraceptive should be used within ____ hours of unprotected intercourse.

- (1) 48
(2) 72
(3) 36
(4) 86

20) Which contraceptive method works by inhibiting ovulation?

- (1) Condom
(2) Oral contraceptive pill
(3) Copper-T
(4) Diaphragm

21) Which part of female genitalia is covered by a condom?

- (1) Mons pubis
(2) Cervix
(3) Labia minora
(4) Clitoris

22) A contraceptive pill contains ____.

- (1) Progesterone and estrogen

- (2) Oxytocin
- (3) Relaxin
- (4) None of these

23) **Assertion** : Lactational amenorrhea have almost nil side effects.

Reason : No medicines or devices are used in this method.

- (1) Assertion & Reason are true and the reason is the correct explanation of the assertion
- (2) Assertion & Reason are true but the reason is not the correct explanation of the assertion
- (3) Assertion is true statement but Reason is false
- (4) Assertion and Reason are false statements

24) India was amongst the first country in the world to initiate action plans and programmes at a national level to attain total reproductive health as a social goal. This programme named as were initiated in.....

- (1) RCH, 1997
- (2) RCH, 1951
- (3) Family planning, 1997
- (4) Family planning, 1951

25) Match the following and choose the correct options :

Column-I		Column-II	
A.	Trophoblast	i.	Embedding of blastocyst in the endometrium
B.	Cleavage	ii.	Group of cells that would differentiate as embryo
C.	Inner cell mass	iii.	Outer layer of blastocyst attached to the endometrium
D.	Implantation	iv.	Mitotic division of zygote

Options :

- (1) A-ii, B-i, C-iii, D-iv
- (2) A-iii, B-iv, C-ii, D-i
- (3) A-iii, B-i, C-ii, D-iv
- (4) A-ii, B-iv, C-iii, D-i

26) Which of the following hormones is not secreted by human placenta?

- (1) hCG
- (2) Estrogen
- (3) Progesterone
- (4) LH

27) Morula is a developmental stage that occur between:

- (1) Between the zygote and blastocyst
- (2) Between the blastocyst and gastrula
- (3) After the implantation
- (4) Between implantation and parturition

28) Assertion: Cu-T and Cu-7 do not suppress sperm motility.

Reason: Hormones released by them affect sperm motility.

- (1) If both Assertion and Reason are true and the Reason is the correct explanation of the Assertion.
- (2) If both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion.
- (3) If Assertion is true but Reason is false.
- (4) If both Assertion and Reason are false

29) **Foetal Ejection Reflex** triggers the release of

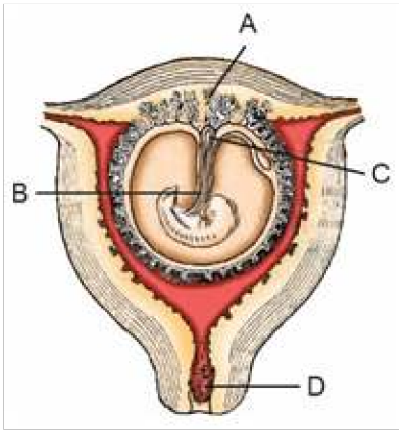
- (1) Oxytocin from foetal pituitary
- (2) Oxytocin from placenta
- (3) Oxytocin from maternal pituitary
- (4) hPL from placenta

30) Match column-I with column-II w.r.t. foetal development in human beings and choose the correct option

	Column - I		Column - II
a.	Foetus develops limbs and digits	i.	End of 24 weeks
b.	External genital organs are well developed	ii.	End of the second month
c.	Hair on head	iii.	End of 12 weeks
d.	Body is covered with fine hair	iv.	Fifth month

- (1) a-(iii), b-(ii), c-(iv), d-(i)
- (2) a-(ii), b-(iii), c-(i), d-(iv)
- (3) a-(ii), b-(iii), c-(iv), d-(i)
- (4) a-(iv), b-(i), c-(ii), d-(iii)

31) In the figure shown below, which labelled structure has **not** been correctly identified?



- (1) A-Placental villi
- (2) B-umbilical cord
- (3) C-Yolk sac
- (4) D-Cervical plug

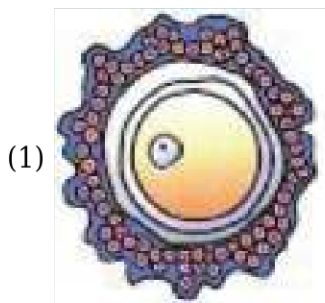
32) Parturition is induced by a complex _____ mechanism. The signals for parturition originate from the _____ and the placenta which induce mild uterine contractions called foetal ejection reflex. Choose the **correct** option to fill in the blanks respectively.

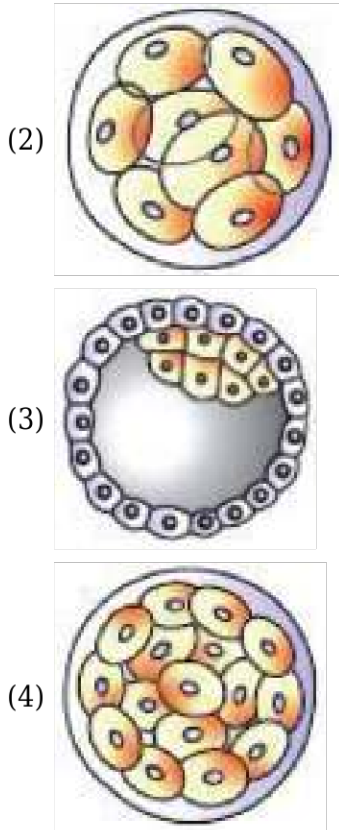
- (1) Neural, maternal pituitary
- (2) Neuroendocrine, fully developed foetus
- (3) Endocrine, foetal pituitary
- (4) Neuroendocrine, half developed foetus

33) Name the hormone secreted by the ovary in the later phase of pregnancy.

- (1) Oxytocin
- (2) Thyroxine
- (3) Cortisol
- (4) Relaxin

34) Which among the following structures is involved in implantation?





35) The process of delivery of foetus is called

- (1) Parturition
- (2) Gestation
- (3) Ejaculation
- (4) Capacitation

SECTION-2

1) Find the incorrect statement about vasectomy.

- (1) It is sterilisation procedure for male
- (2) Vasa efferentia are cut and tied up
- (3) The two vasa deferens are cut and tied up
- (4) Passage of sperm is blocked

2) GIFT technique is recommended for females where:

- (1) Suitable environment for fertilization is not present
- (2) Foetus can't be retained inside uterus
- (3) Cervical canal is too narrow to allow passage for the sperms
- (4) Ovum can't be produced

3) According to MTP (Amendment) Act, 2017, a pregnancy may be terminated on certain considered grounds within

- (1) 12 weeks of pregnancy on the opinion of two registered medical practitioner.
- (2) 20 weeks of pregnancy on the opinion of two registered medical practitioner.
- (3) More than 12 weeks but fewer than 24 weeks, on the opinion of two registered medical practitioner.
- (4) 20 weeks of pregnancy on the opinion of one registered medical practitioner.

4) Which of the following sets involves only *invitro* fertilization methods ?

- (1) ZIFT, IUT
- (2) GIFT, IUT, ICSI
- (3) GIFT, ZIFT, IUT
- (4) GIFT, ICSI

5) According to World Health Organisation, a reproductively healthy individual possesses

- (1) Functionally and Physically normal reproductive organs
- (2) Normal emotional and behavioural interactions with society
- (3) Total well-being in all aspects of reproductions
- (4) All of these

6) The type of natural method of birth control in which the male partner withdrawl his copulatory organ from the vagina just before ejaculation to avoid insemination is :

- (1) withdrawl method
- (2) Lactational Amenorrhea
- (3) Periodic abstinence
- (4) Vesectomy

7) Which of the following is an ideal contraceptive for the females who want to delay pregnancy and/or space children :-

- (1) Barrier method
- (2) Intra uterine devices
- (3) Oral contraceptive pill
- (4) Surgical method

8) Which of the following cannot be detected in developing foetus by amniocentesis ?

- (1) Jaundice
- (2) Haemophilia
- (3) Sex of foetus
- (4) Down syndrome

9) What do you mean induced abortion ?

- (1) Voluntary termination of pregnancy after full terms

- (2) Voluntary termination of pregnancy before full terms
- (3) Foetus removed at incomplete gestation months
- (4) Accidentally foetus is lost

10) **Statement-I :-** In second trimester, abortion are much more riskier.

Statement-II:- Government of india legalised MTP in 1971 with some strict conditions to avoid its misuses.

- (1) Statement I and II both are correct.
- (2) Statement I and II both are incorrect.
- (3) Only Statement I is correct
- (4) Only Statement II is correct

11) Consider the following statements:

- (a) Cervical caps and vaults are not cost effective as they cannot be reused.
- (b) Contraceptive efficiency of vaults and cervical caps can be increased by using spermicidal creams along with these barriers.

Select the correct option

- (1) a is true, b is false
- (2) Both a and b are true
- (3) a is false, b is true
- (4) Both a and b are false

12) What is true for an ideal contraceptive :

- I. It should be user - friendly
- II. It should be easily available
- III. It should be effective and with least side effect
- IV. It should interfere with sexual act of the user.

- (1) Only I, II
- (2) I, II, III
- (3) I, II , IV
- (4) I, II, III, IV

13) Nearly ____ MTPs are performed in a year all over the world.

- (1) 45 to 50 billion
- (2) 45 to 50 million
- (3) 45 to 50 thousands
- (4) 45 to 50 lacs

14) After implantation of blastocyst, some finger like projections appear on the surface of trophoblast called :-

- (1) Yolksac villi
- (2) Chorionic villi

(3) Cilia

(4) Stereocilia

15) Read the following statements and choose the correct option.

Statement - A: A rapid decline in death rate, MMR, and IMR as well as an increase in the number of people in the reproductive age are probable reasons for the population explosion.

Statement-B: Indian population was approximately 350 million at the time of independence.

(1) Statement A is correct but B is incorrect.

(2) Statement A is incorrect but B is correct.

(3) Both statements are correct.

(4) Both statements are incorrect.

PHYSICS

SECTION-1

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	1	1	3	4	1	1	3	3	1	2	2	2	3	2	1	2	1	2	1
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35					
A.	2	1	3	4	1	2	2	4	1	3	3	1	2	1	1					

SECTION-2

Q.	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50
A.	2	1	1	3	3	1	2	4	3	3	1	2	4	1	2

CHEMISTRY

SECTION-1

Q.	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70
A.	2	2	2	2	2	2	3	2	4	2	1	1	1	1	2	4	1	2	2	2
Q.	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85					
A.	2	2	2	1	3	4	4	3	3	2	4	3	3	3	3					

SECTION-2

Q.	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
A.	3	1	2	4	3	3	4	3	1	2	3	2	2	1	2

BOTANY

SECTION-1

Q.	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	4	4	3	3	1	4	3	2	3	2	2	4	4	3	4	2	2	2	2	2
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135					
A.	1	4	2	2	3	1	3	4	3	1	1	4	4	4	3					

SECTION-2

Q.	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	3	2	2	4	3	1	2	2	1	3	3	4	2	1

ZOOLOGY

SECTION-1

Telegram: @NEETxNOAH

Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	3	3	2	1	2	4	3	4	4	4	4	1	4	3	2	3	2	1	2	2
Q.	171	172	173	174	175	176	177	178	179	180	181	182	183	184	185					
A.	2	1	1	4	2	4	1	4	3	3	3	2	4	3	1					

SECTION-2

Q.	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200
A.	2	4	3	1	4	1	2	1	2	1	3	2	2	2	3

PHYSICS

$$8) \quad e = -L \frac{di}{dt}$$

Given, $N_p N_s = 1:10$ and $V_s = 0.5 \times 200 = 100 \text{ V}$
 Now, $\frac{V_s}{V_p} = \frac{N_s}{N_p} \Rightarrow \frac{100}{V_p} = \frac{10}{1} \Rightarrow V_p = 10 \text{ V}$
 $\therefore \frac{i_p}{i_s} = \frac{N_s}{N_p} \Rightarrow \frac{i_p}{0.5} = \frac{10}{1} \Rightarrow i_p = 5 \text{ A}$

9)

11) Length of the rod between the axis of rotation and one of the rod = $\frac{1}{2}$

$$= \pi \left(\frac{\ell}{2} \right)^2 = \left(\frac{\pi \ell^2}{4} \right)$$

Area swept out in one rotation

Angular velocity = $\omega \text{ rad s}^{-1}$

Frequency of revolution = $\frac{\omega}{2\pi}$

Area swept out per second = $\frac{\pi \ell^2}{4} \left(\frac{\omega}{2\pi} \right) = \frac{\ell^2 \omega}{8}$

Magnetic induction = B

Rate of change of magnetic flux = $\left(\frac{B \ell^2 \omega}{8} \right)$

Magnitude of induced emf = $\frac{B \ell^2 \omega}{8}$

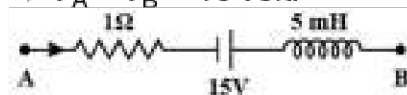
Magnitude of induced emf between the axis and the other end is also $\left(\frac{B \ell^2 \omega}{8} \right)$.

These two emfs are in opposite directions. So, the potential difference between the two ends of the rod is zero.

12) Using Kirchhoff's voltage law,

$$V_A - IR + E - L \frac{di}{dt} = V_B$$

$$\Rightarrow V_A - V_B = 15 \text{ Volt.}$$



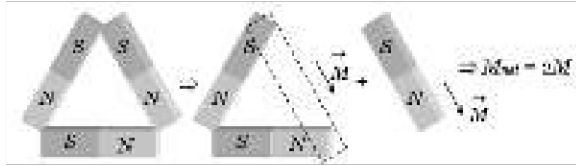
$$\phi = \frac{1}{2} B \omega r^2$$

$$\Rightarrow \phi = \frac{1}{2} \times 0.1 \times 2\pi \times 10 \times (0.1)^2$$

$$\Rightarrow \phi = \pi \times 10^{-2} \text{ V}$$

13)

14) The resultant magnetic moment may be calculated as given below.



$$17) B_1 = \frac{\mu_0 I_1 \alpha_1}{4\pi R}; B_2 = \frac{\mu_0 I_2 \alpha_2}{4\pi R}$$

$$\therefore I_1 \alpha_1 = I_2 \alpha_2$$

$$\frac{B_1}{B_2} = 1$$

So,

$$18) \text{Initially } B_0 = \frac{\mu_0 I}{2R} = 8\sqrt{2}T,$$

$$\text{Finally } B'_0 = \sqrt{\left(\frac{B_0}{2}\right)^2 + \left(\frac{B_0}{2}\right)^2} = \frac{B_0}{\sqrt{2}} = 8T$$

$$19) I = \frac{M}{V} = \frac{M}{\text{mass/density}}$$

$$\text{given mass} = 1\text{gm} = 10^{-3}\text{kg},$$

$$5\text{gm/cm}^3 = \frac{5 \times 10^{-3}\text{kg}}{(10^{-2})^3\text{m}^3}$$

$$\text{and density} = 5 \times 10^3\text{kg/m}^3$$

$$\text{Hence } I = \frac{6 \times 10^{-7} \times 5 \times 10^3}{10^{-3}} = 3$$

20) (A) Due to current carrying wire, the magnetic field in loop will be inwards the paper. As current is increased, magnetic flux associated with loop increases. So a current will be induced so as to decrease magnetic flux inside the loop. Hence induced current in the loop will be anticlockwise. The current in left side of loop shall be downwards and hence repelled by wire. The current in right side of loop is upwards and is hence attracted by wire. Since left side of loop is nearer to wire, repulsive force will dominate. Hence wire will repel the loop.

(B) Options in (B) will be opposite of that in (A).

(C) When the loop is moved away from wire magnetic flux decreases in the loop. Hence the options for this case shall be same as in (B).

(D) When the loop is moved towards the wire magnetic flux increases in the loop. Hence the options for this case shall be same as in (A).

$$21) \phi = (t^3 - 2t) \times 10^{-3} \times 100$$

$$\frac{d\phi}{dt} = (3t^2 - 2) \times 10^{-3} \times 100$$

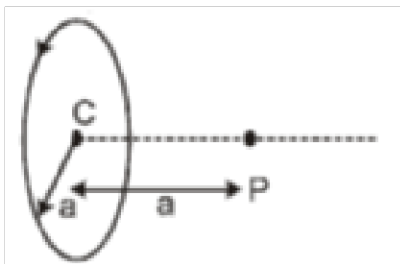
$$\varepsilon = -\frac{d\phi}{dt} = -(3 \times 4 - 2) \times 10^{-3} \times 100 = -1 \text{ V.}$$

24) Here $X_L = \omega L = 2\pi fL$
 $= 2\pi \times 50 \times \frac{0.4}{\pi} = 40 \Omega$
 $R = 30 \Omega$
 $\therefore Z = \sqrt{R^2 + X_L^2} = \sqrt{30^2 + 40^2} = 50 \Omega$
 $I_{rms} = \frac{V_{rms}}{Z} = \frac{200}{50} = 4A.$

25) As $V = 5 \cos(2\pi ft) = 5 \sin(2\pi ft + \pi/2)$
 And $I = 2 \sin(2\pi ft)$
 $\therefore$ Phase difference between V and I is $\phi = \frac{\pi}{2}$
 Average power $P = V_{rms} I_{rms} \times \cos \phi = 0.$

26) $\tan \phi = \frac{X_L}{R} = \frac{2\pi fL}{R} = \frac{2\pi (200/\pi) 1}{200} = 2$
 $\therefore \phi = 63^\circ.$

27) $\omega = \frac{1}{\sqrt{LC}} Q = \frac{L\omega}{R} = \frac{L}{R\sqrt{LC}}$
 $= \frac{4 \times 10^{-3}}{100 \sqrt{4 \times 10^{-3} \times 40 \times 10^{-12}}} = 100.$



28) $\frac{B_C}{B_P} = \frac{\mu_0 i / 2a}{\mu_0 i a^2 / 2(a^2 + a^2)^{3/2}} = \frac{2\sqrt{2}}{1}$

30) $\tau = N I A B \sin \theta$
 $\tau = 50 \times 2 \times 12 \times 10^{-3} \times 0.2 \times \frac{\sqrt{3}}{2}$
 $\tau = 0.20 \text{ Nm}$

31) $X \propto \frac{1}{T} \Rightarrow \frac{X_1}{X_2} = \frac{T_2}{T_1} \Rightarrow \frac{0.0075}{X_2} = \frac{100}{200}$
 $\Rightarrow X_2 = 0.015$

32) $R = \frac{1}{B} \sqrt{\frac{2mV}{1}}$
 $R \propto \sqrt{\frac{m}{q}}$

$$R_p : R_n : R_\alpha = \sqrt{\frac{m}{e}} : \sqrt{\frac{2m}{e}} : \sqrt{\frac{4m}{2e}} \\ = 1 : \sqrt{2} : \sqrt{2}$$

34) Radio = 10^{-2} m (ii)

Micro = 10^{-6} m (iii)

IF = 10^{-4} m

X ray = 10^{-10} m 10^{-4} m (iv)

36) $\frac{1}{2} \epsilon_0 E^2 = \frac{B^2}{2\mu_0}$

$\therefore E$ CB and $C = \frac{1}{\mu_0 \epsilon_0}$

39) Phase difference

$$\Delta\varphi = \varphi_2 - \varphi_1 = \frac{\pi}{6} - \left(-\frac{\pi}{6}\right) = \frac{\pi}{3}$$

41) When ferro mag. material is inserted

$B \uparrow$ so $\phi_B \uparrow$ and $L \propto \mu_r$.

The current in the inner coil, $i = \frac{e}{R} = \frac{A_1}{R_1} \frac{dB}{dt}$,

Length of the inner coil = $2\pi a$

$\therefore$ It's resistance, $R_1 = 50 \times 10^{-3} \times 2\pi(a)$

$$\therefore i_1 = \frac{\pi a^2}{50 \times 10^{-3} \times 2\pi(a)} \times 0.1 \times 10^{-3} = 10^{-4} \text{ A}$$

From the Lenz's law direction of i_1 is clockwise.

Induced current in outer coil, $i_2 = \frac{e_2}{R_2} = \frac{A_2}{R_2} \frac{dB}{dt}$

$$\Rightarrow i_2 = \frac{\pi b^2}{50 \times 10^{-3} \times (2\pi b)} \times 0.1 \times 10^{-3} = 2 \times 10^{-4} \text{ A (CW)}$$

43)

45) Magnetic repulsive force balances the weight.

$$\frac{\mu_0 I_1 I_2 \ell}{2\pi h} = mg \Rightarrow h = \frac{\mu_0 I_1 I_2 \ell}{2\pi mg}$$

$$M_1 = \frac{2}{\pi} \left(\frac{M}{2} \right), M_2 = \frac{M}{2}$$

$$M^1 = \sqrt{M_1^2 + M_2^2}$$

47)

48) Theoretical

49) Compare the given equation with

$$E = E_0 \cos(kz - \omega t)$$

We get, $\omega = 6 \times 10^8 \text{ s}^{-1}$

□ Wave vector, $k = \frac{\omega}{c} = \frac{6 \times 10^8 \text{ s}^{-1}}{3 \times 10^8 \text{ ms}^{-1}} = 2 \text{ m}^{-1}$

CHEMISTRY

57) $\Delta H = E_f - E_b$

$-200 = 80 - E_b$

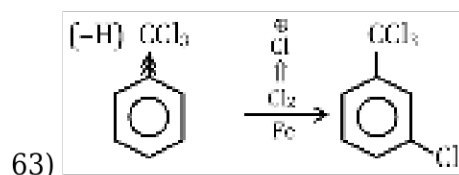
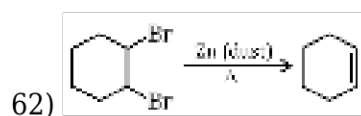
$E_b = 280 \text{ kJ/mol}$

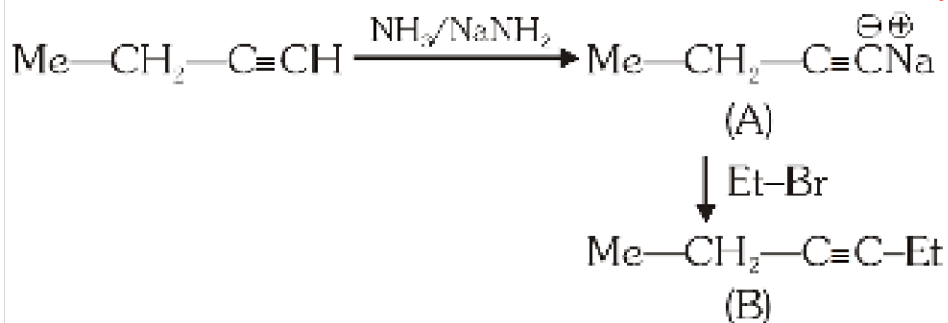
58) $\text{Rate} = k[\text{B}]^2$

61)

[Boiling point $\propto$ molecular weight]

$\text{CH}_3\text{—CH}_2\text{—I}$ is having maximum amongst all.

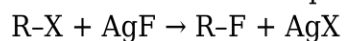




67)

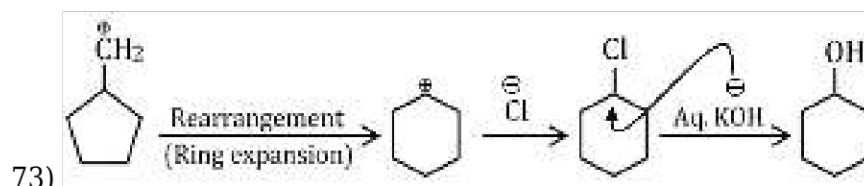
70)

Swarts reaction is for preparation of alkyl fluoride by using other alkyl halide.

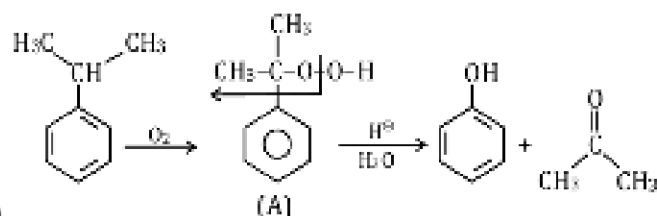


(X = Cl, Br)

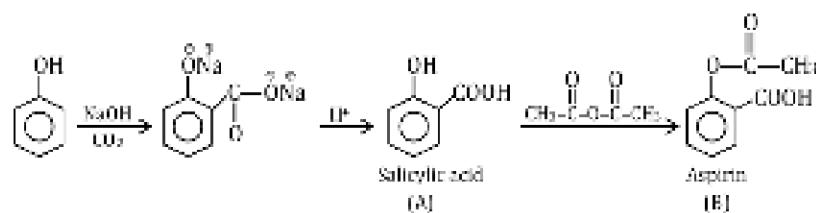
71) NCERT Reference: XII, Part-II, Page No. 335



74)



75)



79) Phenol is more reactive than benzene towards electrophilic substitution reaction due OH group is ring activator

BOTANY

101)

133) NCERT Pg. # 18

149)

NCERT Pg. No. 26 Fig 2.6 (a)

ZOOLOGY

161) NCERT-XII Pg#48

162) NCERT Pg # 46

164)

NEW NCERT pg. # 48

165)

NEW NCERT pg. # 42

166)

NEW NCERT pg. # 47

167) Ncert Pg No: 44

168) Ncert Pg No: 44,45

179) NCERT Reference: Biology-XII, Page No.- 54

181) NCERT Reference: Biology-XII, Page No.- 53

183) Ncert Pg No: 37

184) Ncert page 36.

185) Ncert Pg No: 38

190) NCERT-XII, Pg. # 57

195) NCERT, Pg. # 46 (E)

196)

NCERT Pg - 60

200) Ncert Pg No . 59